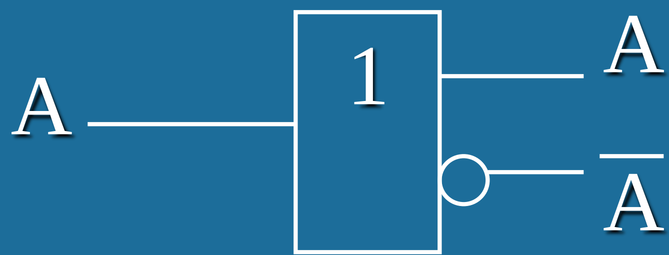
Chapter 8 Memory and Programmable Logic Device

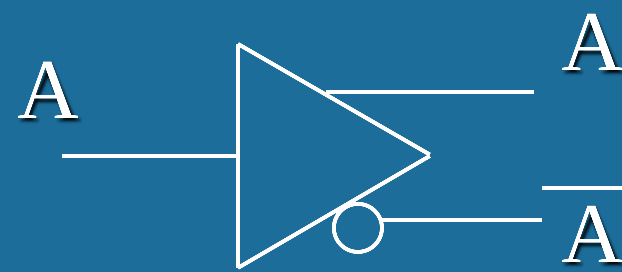
- 8.1 Programmable Logic Device
- 8.2 Read-Only Memory
- 8.3 Random-Access Memory

8.1 Programmable Logic Device (PLD)

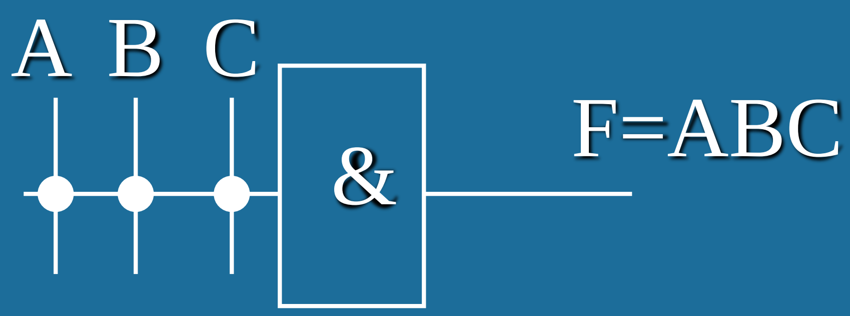
PLD	AND array	OR array	Output circuit
PROM	Fixed	Programmable	Fixed
PLA	Programmable	Programmable	Fixed
PAL	Programmable	Fixed	Fixed
GAL	Programmable	Fixed	Configurable

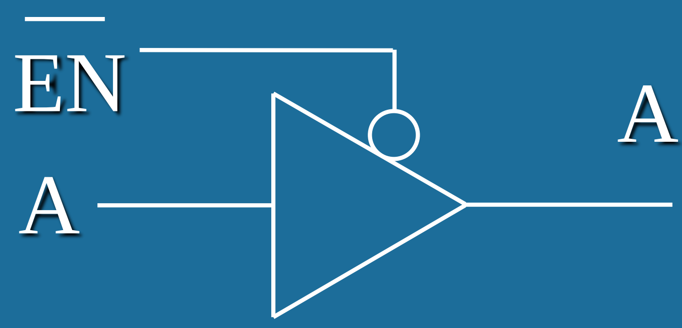
(1) PLD Buffer





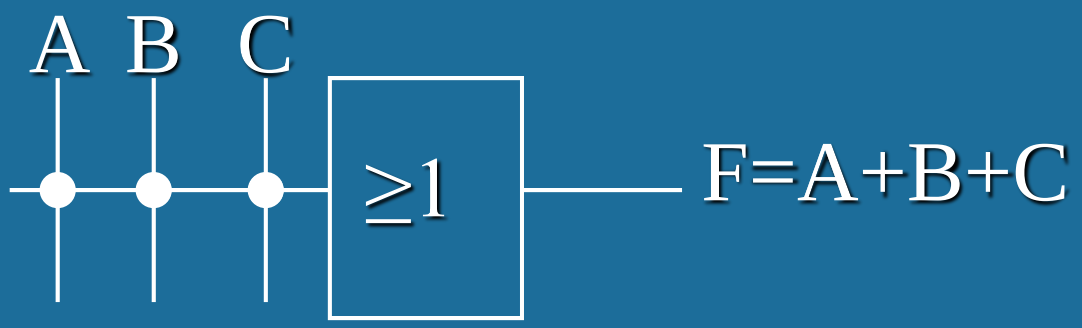
(2) PLD AND Gate





Tri-State Buffer

(3) PLD OR Gate



(4) PLD Connection

Hardware connection

Programmed connection

Disconnected

8.2 Read-Only Memory (ROM)

Type:

ROM ——**BIOS**

RAM { **SRAM（Static RAM）** ——**Cache**
　　 DRAM（Dynamic RAM） —— **memory**

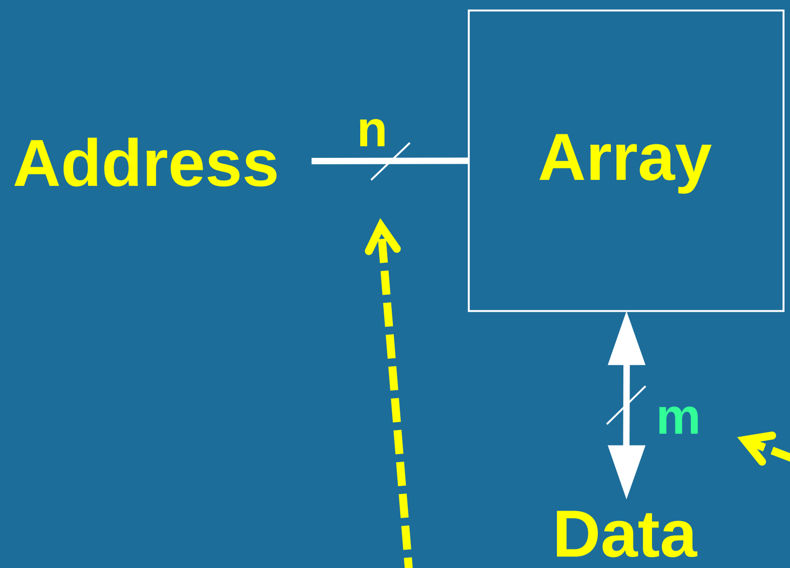
Specs:

- **Capacity：words ×bits**

  （example：8K×1bit，256M×8bit =256MB ）

- **Speed： read/write cycle**

1. Basic structure： n-bit address input、 m-bit output



- ROM capacity：

n inputs→2^n addresss→2^n memory units→capacity: $2^n \times m$

64MB memory， has __26__ bit address.

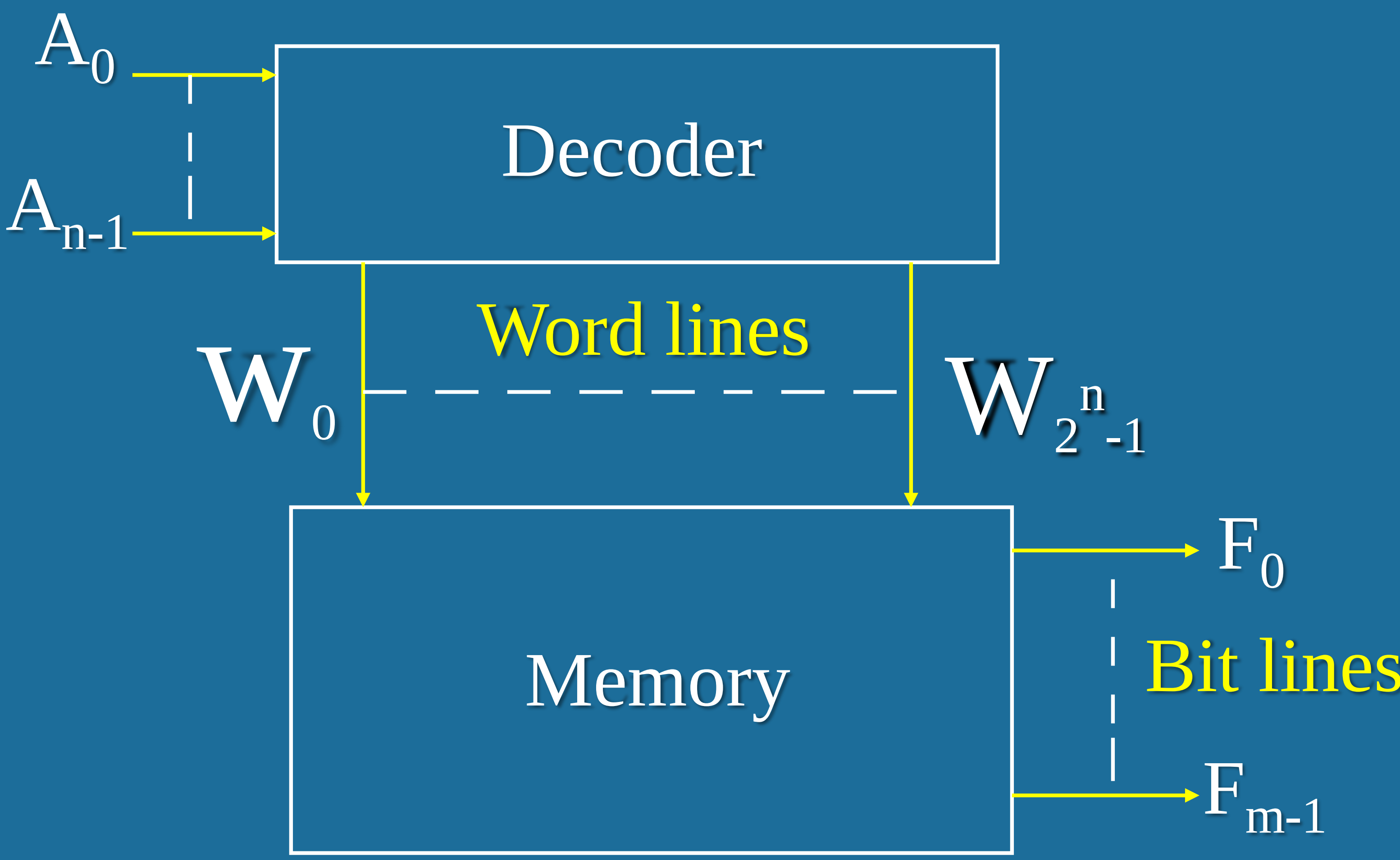
Suppose that the capacity of ROM is “64MB”. Calculate the number of input variables of ROM.

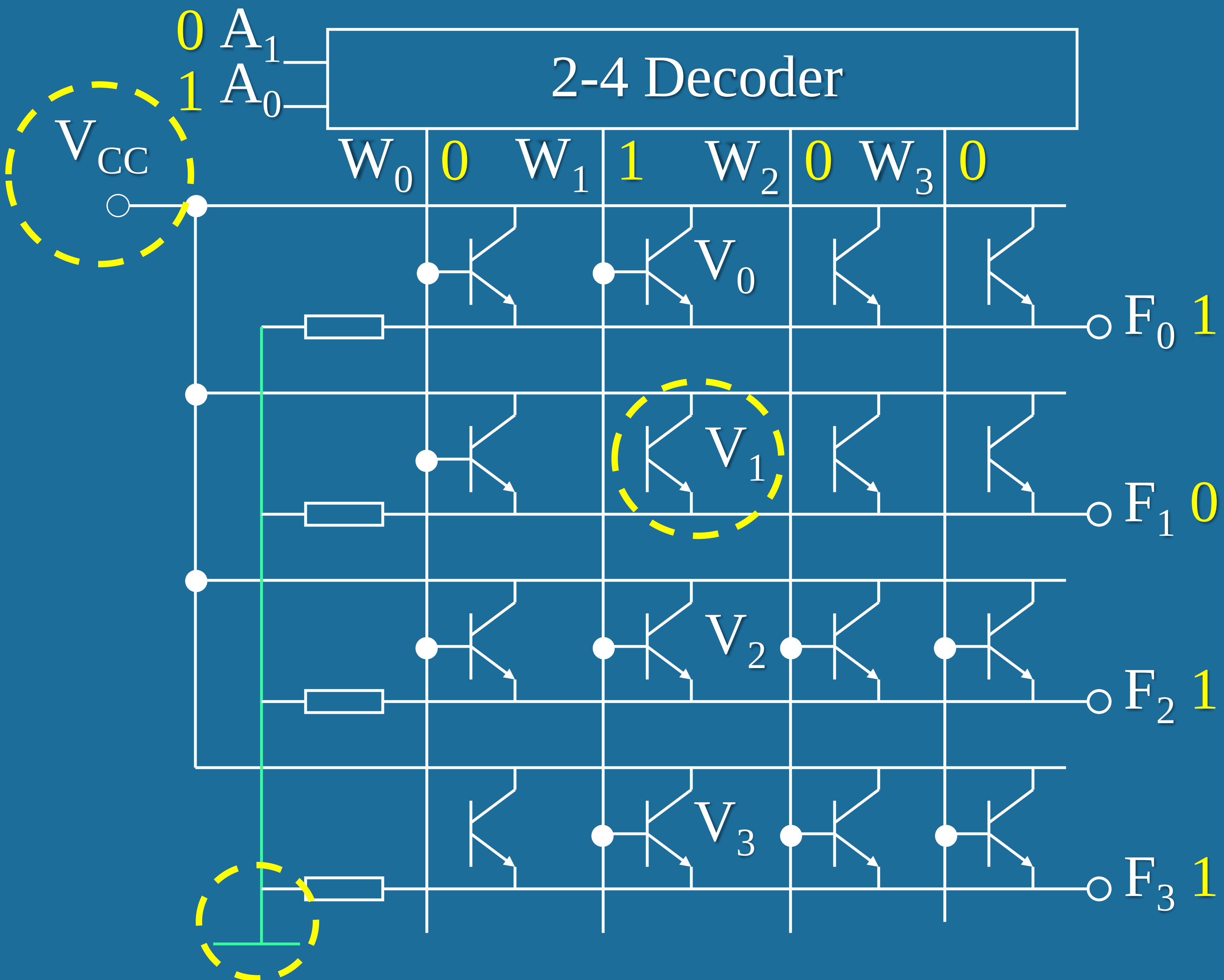
ROM Capacity	Number of Word Lines	Number of Bit Lines
64MB	64M bit	1 B = 8 bit

Address lines to Word Lines is a 26-64M decoder.

Number of Word Lines	AND array	
	Number of Output Variables	Number of Input Variables
64M bit	$64\text{M} = 64 \times 1024 \times 1024$ $= 2^6 \times 2^{10} \times 2^{10} = 2^{26} = 2^n$	$n = 26$

ROM with capacity $2^n \times m$ bit





0 A1
1 A0
2-4 Decoder
VCC
W0 0
W1 1
W2 0
W3 0
V0
V1
V2
V3
F0 1
F1 0
F2 1
F3 1

Logic functions

$F_0 = W_0 + W_1$

$F_1 = W_0$

$F_2 = W_0 + W_1 + W_2 + W_3$

$F_3 = W_1 + W_2 + W_3$

$$W_0 = \overline{A_1} \cdot \overline{A_0} \qquad W_1 = \overline{A_1} A_0 \qquad W_2 = A_1 \overline{A_0} \qquad W_3 = A_1 A_0$$

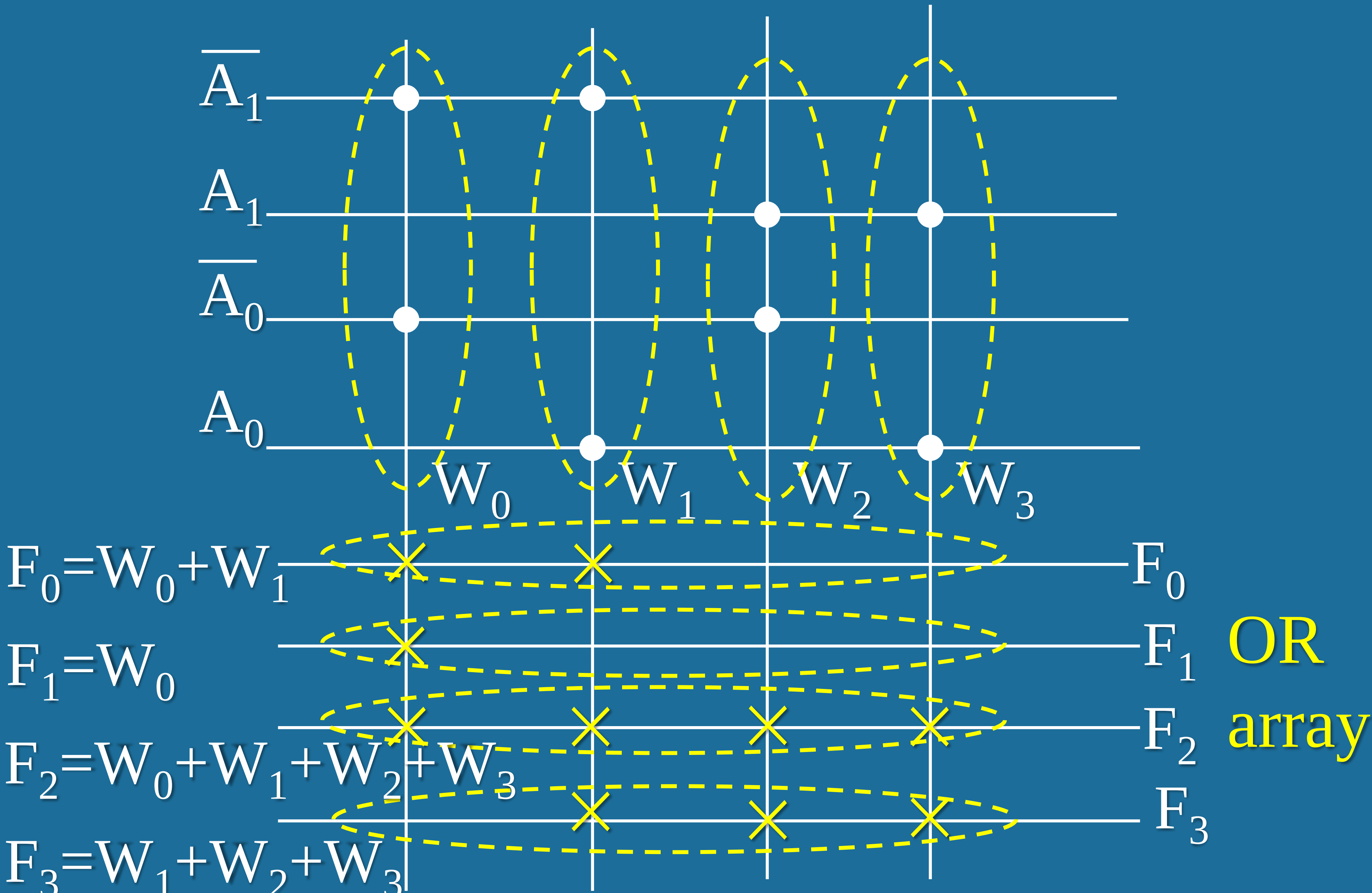
$$F_0 = \overline{A_1} \cdot \overline{A_0} + \overline{A_1} A_0$$

$$F_1 = \overline{A_1} \cdot \overline{A_0}$$

$$F_2 = \overline{A_1} \cdot \overline{A_0} + \overline{A_1} A_0 + A_1 \overline{A_0} + A_1 A_0$$

$$F_3 = \overline{A_1} A_0 + A_1 \overline{A_0} + A_1 A_0$$

AND array



$\overline{A_1}$
A_1
$\overline{A_0}$
A_0
W_0
W_1
W_2
W_3
$F_0=W_0+W_1$
$F_1=W_0$
$F_2=W_0+W_1+W_2+W_3$
$F_3=W_1+W_2+W_3$
F_0
F_1
F_2
F_3
OR
array

Implement the following logic functions by the "AND-OR" array of ROM.

$$F_1 = \overline{A} \cdot \overline{C} + AB\overline{C}$$

$$F_2 = A\overline{B} + ABC$$

$$F_3 = AC + \overline{A}B$$

$$F_1 = \overline{A}\cdot\overline{C}(B+\overline{B}) + AB\overline{C} = \overline{A}\cdot\overline{B}\cdot\overline{C} + \overline{A}B\overline{C} + AB\overline{C}$$

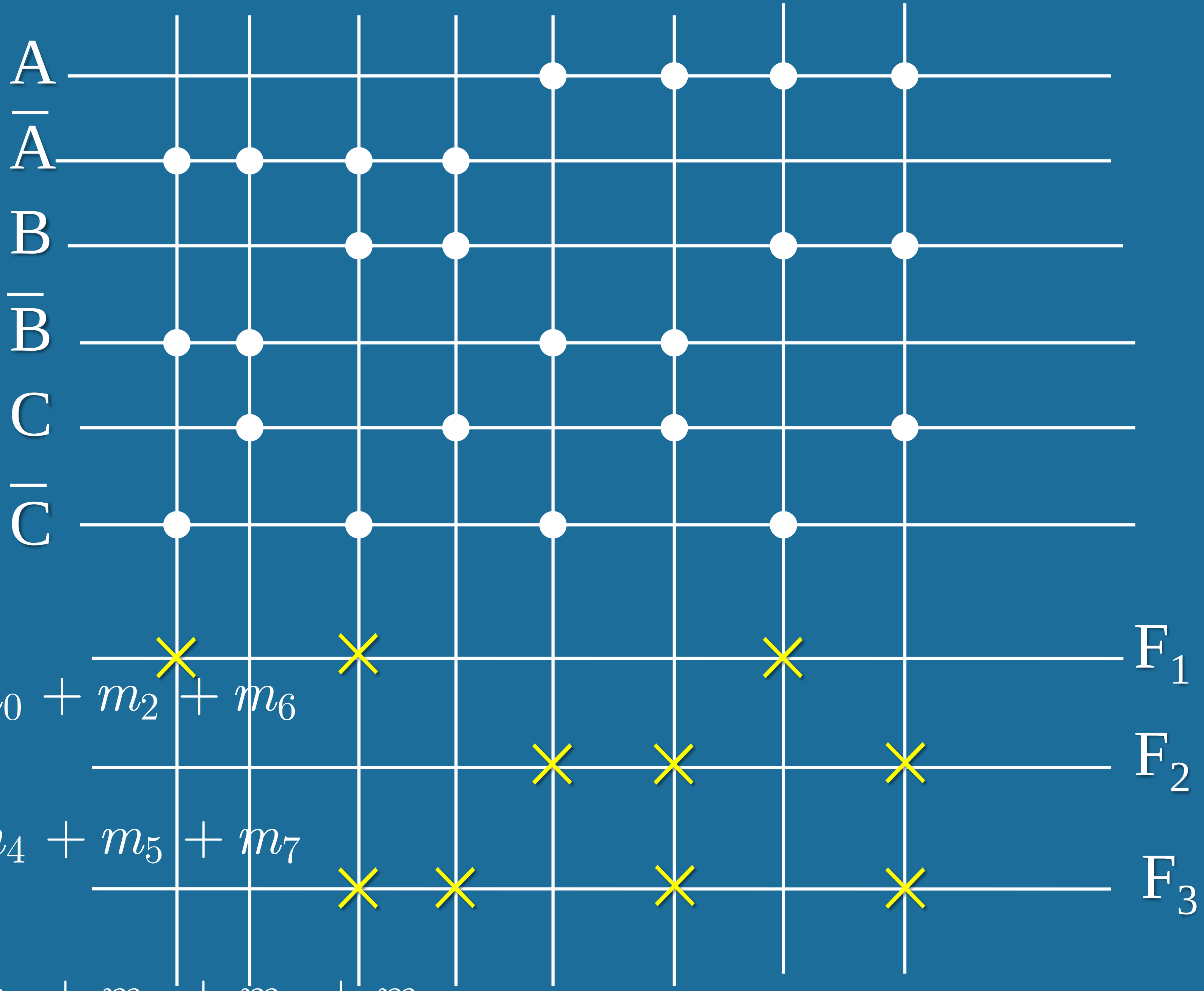
$$= m_0 + m_2 + m_6$$

$$F_2 = A\overline{B}(C+\overline{C}) + ABC = A\overline{B}\cdot\overline{C} + A\overline{B}C + ABC$$

$$= m_4 + m_5 + m_7$$

$$F_3 = AC(B+\overline{B}) + \overline{A}B(C+\overline{C}) = A\overline{B}C + ABC + \overline{A}B\overline{C} + \overline{A}BC$$

$$= m_2 + m_3 + m_5 + m_7$$



$$F_1 = m_0 + m_2 + m_6$$

$$F_2 = m_4 + m_5 + m_7$$

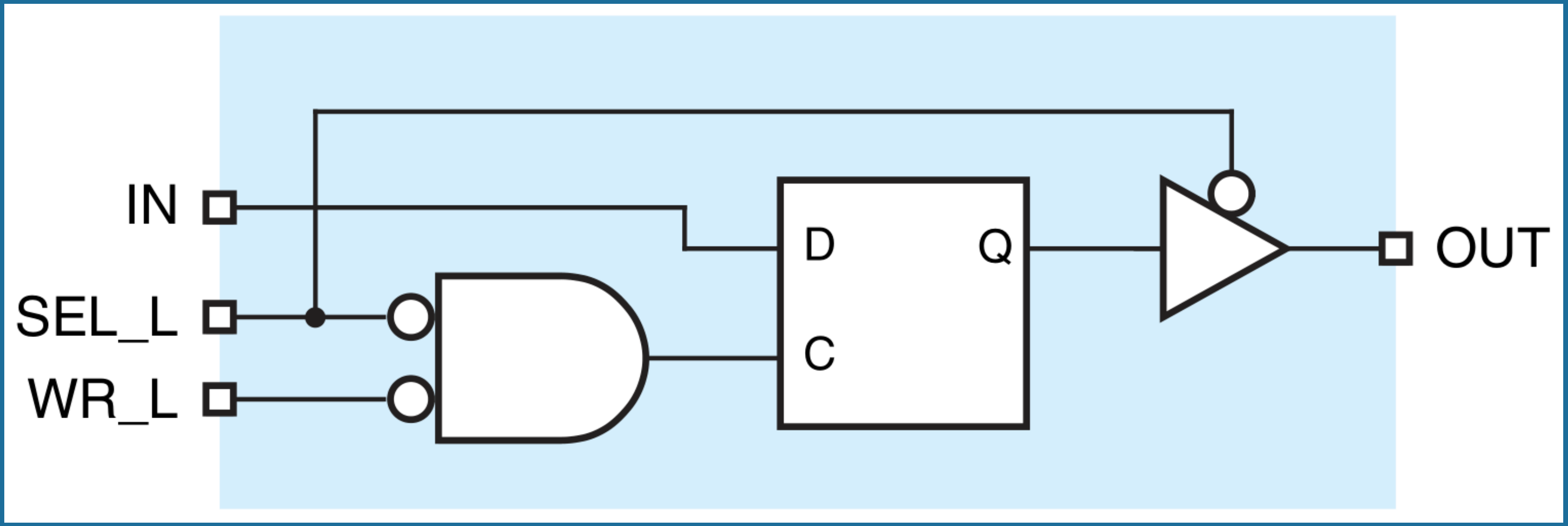
$$F_3 = m_2 + m_3 + m_5 + m_7$$

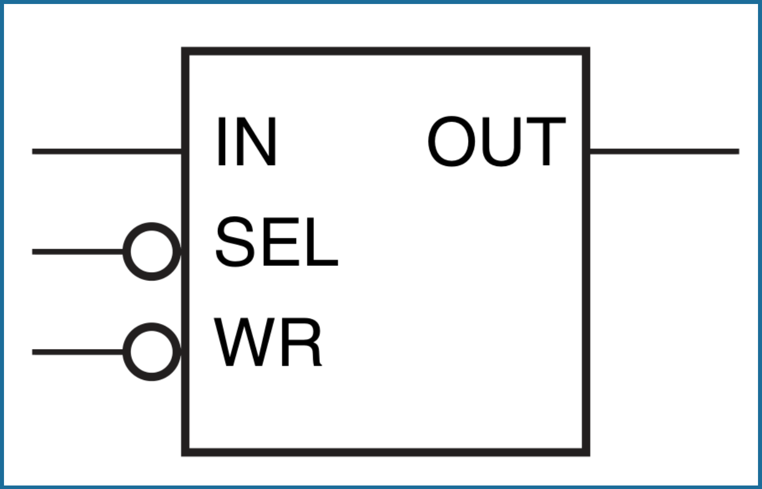
8.3 Random-Access Memory

(1) Static Random-Access Memory (SRAM)

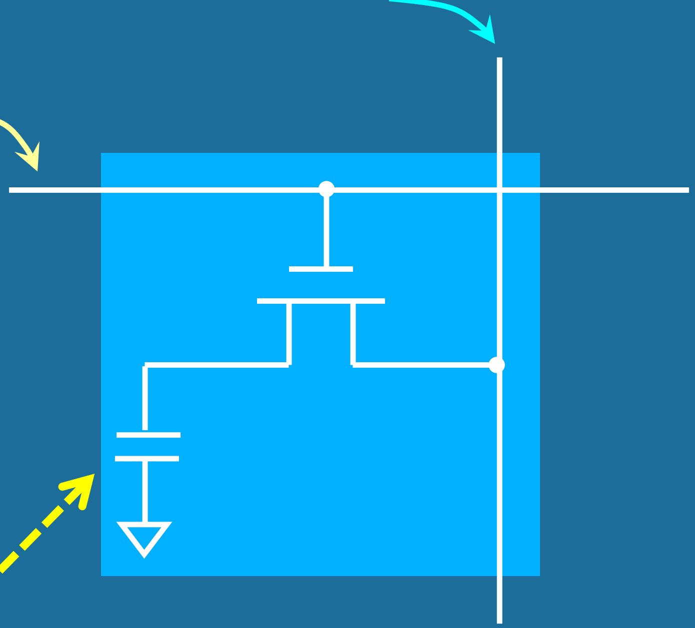
(2) Dynamic Random-Access Memory (DRAM)

SRAM composed of D flip-flop and buffer circuits





DRAM composed of one NMOS transistor and one capacitor



Word line high state →
“gate” high state→
NMOS transistor “ON” →
DRAM selected

Bit line high state → capacitor charging → “1” stored on DRAM

Bit line low state → capacitor discharging → “0” stored on DRAM